

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

May 2013

CL 404 Water Resources Engineering & Management

Date: 10.05.2013, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q – 1(a) Enlist different structural measures of flood control and discuss any ONE of them in detail. [04]
- (b) What is drought? Discuss the measures for drought protection. [03]
- Q – 2(a) What are the types of the reservoir? Discuss the factors affecting the selection of a suitable site for the reservoir. [06]
- (b) What are the measures for the control of reservoir sedimentation? [04]
- (c) Explain the importance of the rivers in the development human civilization. Also, distinguish between flashy river and virgin river. [04]

OR

- Q – 2(a) Explain, with neat sketch, various zones of the reservoir. What is the significance of dead storage in a reservoir? [06]
- (b) What is reservoir sedimentation? Also, explain 'density current' and 'trap efficiency'. [04]
- (c) Explain the phenomenon of delta formation in the rivers. [04]
- Q – 3(a) Explain the detailed classification of hydel power scheme. [05]
- (b) Differentiate between Thermal power plant and Hydropower plant. [05]
- (c) Discuss the objectives of river training. [04]

OR

- Q – 3(a) Explain the following terms: [03]
(i) Load factor (ii) Firm power (iii) Installed capacity
- (b) Draw a neat sketch showing different components of a hydro electric project and explain the importance of the surge tank. [07]
- (c) Enlist different methods of river training and explain any ONE of them. [04]

SECTION – II

- Q – 4(a) Define following: [04]
(i) Catchment yield (ii) Reservoir yield (iii) Dependable yield (iv) Safe yield
- (b) How the investigations for reservoir planning are carried out? [03]
- Q - 5 (a) What are the objectives of water resources development? Classify WRD Projects with their functional requirements. How any WRDP is formulated? [07]

- (b) The following table gives the mean monthly flows in a river during 1981. [07]
Calculate the minimum storage required to maintain a demand rate of $40 \text{ m}^3/\text{sec}$ for two months by analytical method.

Month	Jan	Feb	Mar	April	May	June	July	Aug	Sept	Oct	Nov	Dec
Mean Flow, m^3/sec	60	45	35	25	15	22	50	80	105	90	80	70

OR

- Q - 5 (a) Discuss planning strategy for water resources development projects. List out various steps involved to plan WRP. Explain in detail data required to be collected for planning of a WRP. [07]

- (b) Write short note on following: [07]
(i) Reservoir losses (ii) Reservoir clearance

- Q - 6 (a) For better planning and management of project which strategies are carried out? [07]

- (b) A proposed reservoir has a capacity of 450 ha-m and catchment area 250 km^2 . The annual stream flow averages 10 cm of runoff and production of annual sediment is 0.045 ha-m/km^2 . Calculate the annual sediment inflow in the reservoir in Mm^3 . [07]

OR

- Q - 6 (a) Enlist the stages of investigation for WRDP. Explain each one in detail. [07]

- (b) The yearly data for a catchment of a proposed reservoir site for 30 years is given in following table. Compute the dependable rainfall for 55% & 70% dependability. [07]

Year	1945	1946	1947	1948	1949	1950	1951	1952	1953	1954
Rainfall (cm)	78	76	101	112	80	85	116	120	140	150
Year	1955	1956	1957	1958	1959	1960	1961	1962	1963	1964
Rainfall (cm)	99	98	66	70	104	92	95	87	69	91
Year	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974
Rainfall (cm)	131	138	126	128	120	111	110	83	82	94
